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INTRODUCTION

The **pharmacy management system**, also known as the **pharmacy information system**, is a system that stores data and enables functionality that organizes and maintains the medication use process within pharmacies.

Pharmacy management system can make the work easier by giving the details of the medicine when its name is entered. A computer gives the details of the medicine like rate of medicine. It becomes very difficult in big medical stores to handle the details of all the medicines normally, so by using this pharmacy management system we can maintain the records of all the medicines.

The Pharmacy Management System application has more benefits. Easy to find medicine because we use strong search option and easy to add medicine, medicine type, medicine deadline and also delete and modify option its help to both of customer and pharmacist.

OBJECTIVES

The main objective of the project is to let the students apply their knowledge of programming into real life projects. It also helps the students to improve their skills like team management, programming skills. It improve their way of organizing data's.

The main objectives of Pharmacy Management system are:

- Provides the searching facilities based on various factors such as inventory, Expired medicines.
- It shows the information and description of medicines in the inventory.
- It can search through inventory by using filters such as names of the medicine and name of the manufacturer.
- Increases the overall efficiency of managing drugs, capsules, etc.
- It can add, delete or update the records of medicinal data.

WORKING ENVIRONMENT

What is python?

Python is a popular programming language. It was created by Guido Van Rossum and released in 1991.Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python is dynamically-typed and garbage-collected. It supports multiple programming paradigms, including structured, object-oriented and functional programming.

It is used for:

- Web development (server-side)
- Software development
- Data analysis
- System scripting

What Python can do ?

Python can be used on a server to create web applications.

- Python can be used alongside software to create workflows.
- Python can connect to database systems. It can also read and modify files.
- Python can be used to handle big data and perform complete mathematics.
- Python can be used for rapid prototyping, or for production ready software development.

What is MYSQL?

MySQL is an open source relational database management system (RDMS). It is the most popular database system used with PHP. MYSQL is developed, distributed and supported by Oracle Corporation.

- The data in MySQL database are stored in tables which consists of columns and rows.
- MySQL is a database system that runs on a server.
- MySQL is ideal for both small and large applications.
- MySQL is very fast, reliable and easy to use database system. It uses standard SQL.

Interfacing Python with MySQL

MySQL python/connector is an interface for connecting to a mysql database server from python. It implements the python database API and is built on top of the MySQL.

The general workflow of a python program that interacts with a MySQL based database is as follows:

1. Connect to MySQL server.
2. Create a new database.
3. Connect to the newly created or an existing database.
4. Execute a SQL query and fetch results.
5. Inform the database if any changes are made to a table.
6. Close the connection to the MySQL server.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

- Processor : Intel Core i5-1135G7 or above
- Processor speed : 2.30 GHz
- Ram : 4GB or more
- Hard disk : 800 GB or more

Software:

- Operating system : Windows 7 or above
- IDE : IDLE Python
- Front-end : Python 3.6 or above
- Back-end : My SQL server 5.0 or above

PROPOSED SYSTEM

Proposed system is system which is computerized in every manner. Computerized system is not just adding machines but they are capable of doing much complex, tedious and cumbersome tasks.

Processing of data by hand is satisfactorily only when the amount of data to be processed is small and also the Manuel processing is slow, monotonous and often subject to error.

Above explanation is clearly telling us that existing system contains a lot of deficiencies which can be removed by only by following proposed system.

Nowadays, computer graph is at its extent. Computerization contains a lot of benefits so that everyone chasing and following computerized items. Now , question arises what kind of help this project or computerized system can give to remove all disadvantages of this existing system.,

SYSTEM DEVELOPMENT LIFE CYCLE

The systems development life cycle is a project management technique that divides complex projects into smaller, more easily managed segments or phases. Segmenting projects allows managers to verify the successful completion of project phases before allocating resources to subsequent phases.

Software development projects typically include initiation,planning,design,development,testing,implementation,and maintenance phases. However, the phases may be divided differently depending on the organization involved.

For example, initial project activities might be designated as request, requirements-definition, and planning phases, or initiation, concept-development, and planning phases. End users of the system under development should be involved in reviewing the output of each phase to ensure the system is being built to deliver the needed functionality.

PHASES OF SYSTEM DEVELOPMENT LIFE CYCLE

Phase 1: Requirements collection and analysis

In this phase we mainly focus on gathering the business needs from the customer. It determines the requirements like; what should be the input data to the system? Who is going to use the system? What should be the output data by the system?. These questions are getting answered during this phase.

Phase 2: Designing

The design phase involves converting the informational, functional, and network requirements identified during the initiation and planning phases into unified design specifications that developers use to script programs during the development phase. Program designs are constructed in various ways. During this phase, the system is designed to satisfy the functional requirements identified in the previous phase. Since problems in the design phase could be very expensive to solve in the later stage of the software development, a variety of elements are considered in the design to mitigate risk.

Phase 3: Development

The development phase involves converting design specifications into executable programs. Effective development standards include requirements that programmers and other project participants discuss design specifications before programming begins. The procedures help ensure programmers clearly understands program designs and functional requirements. Programmers use various techniques to develop computer programs. Procedural programming involves the line-by-line scripting of logical instructions that are combined to form a program.

Phase 4: Implementation

This phase is initiated after the system has been tested and accepted by the user. In this phase, the system is installed to support the intended business functions. System performance is compared to performance objectives established during the planning phase. Implementation includes user notification, user training, installation of hardware, installation of software onto production computers, and integration of the system into daily work processes. This phase continues until the system is operating in production in accordance with the defined user requirements.

SOURCE CODE

```
import mysql.connector as a
con=a.connect(host='localhost',user='root',password='admin',database="rit
k")

def Add():
                cur_obj=con.cursor()
                ref=int(input('Enter the drug number: '))
                name=input('Enter the name: ')
                manuftr=input('Enter the manuftr name: ')

                manuftr_date=input('Enter the manuftr date(YYYY-MM-DD): ')
                exp_date=input('Enter the expiry date(YYYY-MM-DD): ')
                quantity=int(input('Enter the quantity(no.of
tablets/vaccines/etc.): '))
                category=input('Enter the category: ')
                unitpr=int(input('Enter the unitprice(per lable/per bottle/etc.): '))
                totalpr=int(input('Enter the total price(total stock price): '))

        val=(ref,name,manuftr,manuftr_date,exp_date,quantity,category,unitp
r,totalpr)
                query='insert into INVENTORY
values(%s,%s,%s,%s,%s,%s,%s,%s,%s);'
                cur_obj.execute(query,val)
                con.commit()
```

```
		print('DETAILS ADDDED SUCCESFULLY')
		exit=input('PRESS ENTER TO ENTER THE MAIN MENU')
		print("")
		main()

def delete():
		cur_obj=con.cursor()
		ref=int(input('Enter the drug number: '))
		cur_obj.execute('DELETE FROM INVENTORY WHERE
ref_num={};'.format(ref))
		con.commit()
		print('DETAILS DELETED SUCCESFULLY')
		exit=input('PRESS ENTER TO ENTER THE MAIN MENU')
		print("")
		main()

def update():
		print('Select the name of the column to be updated')
		print('1.Drug name')
		print('2.Quantity')
		print('3.total price')
		ch=int(input('Enter the choice number(1-3):'))
		import mysql.connector as sqltor

	mycon=sqltor.connect(host='localhost',user='root',passwd='admin',dat
abase='ritk')
```

```
        cur_obj=mycon.cursor()
        if ch==1:
            p=input('Enter the new drug name to be updated: ')
            m=int(input('Enter the reference number of the drug'))
            query='update INVENTORY set DrugName="{}"where
ref_num={}'.format(p,m)
            cur_obj.execute(query)
            mycon.commit()
            print('DETAILS UPDATED SUCCESFULLY')
        elif ch==2:
            q=int(input('Enter the new quantity to be updated: '))
            n=int(input('Enter the reference number of the drug: '))
            query='update INVENTORY set Quantity={} where
ref_num={}'.format(q,n)
            cur_obj.execute(query)
            mycon.commit()
            print('DETAILS UPDATED SUCCESFULLY')
        elif ch==3:
            s=int(input('Enter the new total price to be updated: '))
            xy=int(input('Enter the reference number of the drug'))
            query='update INVENTORY set Total_pr={} where
ref_num={}'.format(s,xy)
            cur_obj.execute(query)
            mycon.commit()
            print('DETAILS UPDATED SUCCESFULLY')
        else:
```

```
        print('invalid choice')
    exit=input('PRESS ENTER TO ENTER THE MAIN MENU')
    print("")
    main()

def showall():
    a='select * from INVENTORY;'
    cursor_obj=con.cursor()
    cursor_obj.execute(a)
    data=cursor_obj.fetchall()
    for i in data:
        print('Ref_num: ',i[0])
        print('Drug_name: ',i[1])
        print('Manuftr_name: ',i[2])
        print('Manuftring_date: ',i[3])
        print('Expiry_date: ',i[4])
        print('Quantity: ',i[5])
        print('Category: ',i[6])
        print('Unit_pr: ',i[7])
        print('Total_pr: ',i[8])
        print('')
        print('')
        exit=input('PRESS ENTER TO ENTER THE MAIN
MENU')
        print("")
        main()
```

```
def expired():
   date=input("Enter todays date(YYYY-MM-DD): ")
   c=con.cursor()
   c.execute('select * from INVENTORY where
Expiry_date<"{}";'.format(date))
   data=c.fetchall()
   for i in data:
         print('')
         print('')
         print('Ref_num: ',i[0])
         print('Drug_name: ',i[1])
         print('Manuftr_name: ',i[2])
         print('Manuftring_date: ',i[3])
         print('Expiry_date: ',i[4])
         print('Quantity: ',i[5])
         print('Category: ',i[6])
         print('Unit_pr: ',i[7])
         print('Total_pr: ',i[8])
         print('')
         print('')
         exit=input('PRESS ENTER TO ENTER THE MAIN MENU')
         print("")
         main()
```

```
def main():
    print('==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========')
    print('1.Add new drug details')
    print('2.delete drug details')
    print('3.update details')
    print('4.show expired drugs')
    print('5.show inventory')

    ch=int(input('Enter the choice(1-5): '))
    if ch==1:
        Add()
    elif ch==2:
        delete()
    elif ch==3:
        update()
    elif ch==4:
        expired()
    elif ch==5:
        showall()
    else:
        print('invalid choice')
main()
```

PROGRAM DEMO

- **Table description**

mysql> desc inventory;

Field	Type	Null	Key	Default	Extra
ref_num	int	NO	PRI	NULL	
DrugName	varchar(100)	YES		NULL	
Manufturer	varchar(100)	YES		NULL	
Manuftring_date	date	YES		NULL	
Expiry_date	date	YES		NULL	
Quantity	int	YES		NULL	
Category	varchar(100)	YES		NULL	
Unit_pr	int	YES		NULL	
Total_pr	int	YES		NULL	

9 rows in set (0.13 sec)

- **Table data**

mysql> select * from inventory;

ref_num	DrugName	Manufturer	Manuftring_date	Expiry_date	Quantity	Category	Unit_pr	Total_pr
1	Abilify(aripiprazole)	mankind	2020-12-16	2023-04-15	64	tablet	120	20000
2	dolo	mankind	2020-11-22	2022-10-22	100	tablets	15	50000
3	Covishield	Bharat biotech International Ltd	2020-02-05	2021-02-05	50	vaccine	100	5000
12	paracetamol	crocin	2020-10-15	2022-11-09	100	tablets	10	10000
45	saridon	mankind	2020-06-11	2021-06-12	100	tablet	15	1500

5 rows in set (0.02 sec)

- **Add function**

```
>>>
= RESTART: C:\Users\Student 06\Desktop\pharmacy management system\1672587589336_main_PMS.py
==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========
1.Add new drug details
2.delete drug details
3.update details
4.show expired drugs
5.show inventory
Enter the choice(1-5): 1
Enter the drug number: 33
Enter the name: penicillin
Enter the manuftr name: INJECT CARE PARENTERALS PVT LTD
Enter the manuftr date(YYYY-MM-DD): 2020-07-04
Enter the expiry date(YYYY-MM-DD): 2022-06-04
Enter the quantity(no.of tablets/vaccines/etc.): 100
Enter the category: antibiotics
Enter the unitprice(per lable/per bottle/etc.): 800
Enter the total price(total stock price): 8000
DETAILS ADDDED SUCCESFULLY
PRESS ENTER TO ENTER THE MAIN MENU
```

mysql> select * from inventory;

ref_num	DrugName	Manufturer	Manuftring_date	Expiry_date	Quantity	Category	Unit_pr	Total_pr
1	Abilify(aripiprazole)	mankind	2020-12-16	2023-04-15	64	tablet	120	20000
2	dolo	mankind	2020-11-22	2022-10-22	100	tablets	15	50000
3	Covishield	Bharat biotech International Ltd	2020-02-05	2021-02-05	50	vaccine	100	5000
33	penicillin	INJECT CARE PARENTERALS PVT LTD	2020-07-04	2022-06-04	100	antibiotics	800	8000
45	saridon	mankind	2020-06-11	2021-06-12	100	tablet	15	1500

5 rows in set (0.01 sec)

- **Delete function**

```
==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========
1.Add new drug details
2.delete drug details
3.update details
4.show expired drugs
5.show inventory
Enter the choice(1-5): 2
Enter the drug number: 12
DETAILS DELETED SUCCESFULLY
PRESS ENTER TO ENTER THE MAIN MENU
```

```
mysql> select * from inventory;
+---------+----------------------+-----------------------------+----------------+-------------+----------+-------------+---------+----------+
| ref_num | DrugName             | Manufturer                  | Manuftring_date | Expiry_date | Quantity | Category    | Unit_pr | Total_pr |
+---------+----------------------+-----------------------------+----------------+-------------+----------+-------------+---------+----------+
|       1 | Abilify(aripiprazole) | mankind                    | 2020-12-16     | 2023-04-15  |       64 | tablet      |     120 |    20000 |
|       2 | dolo                 | mankind                     | 2020-11-22     | 2022-10-22  |      100 | tablets     |      15 |    50000 |
|      33 | penicillin           | INJECT CARE PARENTERALS PVT LTD | 2020-07-04 | 2022-06-04  |      100 | antibiotics |     800 |     8000 |
|      45 | saridon              | mankind                     | 2020-06-11     | 2021-06-12  |      100 | tablet      |      15 |     1500 |
+---------+----------------------+-----------------------------+----------------+-------------+----------+-------------+---------+----------+
4 rows in set (0.00 sec)
```

- **Update function**

a. NAME:

```
>>>
= RESTART: C:\Users\Student 06\Desktop\pharmacy management system\1672587589336_main_PMS.py
==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========
1.Add new drug details
2.delete drug details
3.update details
4.show expired drugs
5.show inventory
Enter the choice(1-5): 3
Select the name of the column to be updated
1.Drug name
2.Quantity
3.total price
Enter the choice number(1-3):1
Enter the new drug name to be updated: benedril
Enter the reference number of the drug45
DETAILS UPDATED SUCCESFULLY
PRESS ENTER TO ENTER THE MAIN MENU
```

mysql> select * from inventory;

ref_num	DrugName	Manufturer	Manuftring_date	Expiry_date	Quantity	Category	Unit_pr	Total_pr
1	Abilify(aripiprazole)	mankind	2020-12-16	2023-04-15	64	tablet	120	20000
2	dolo	mankind	2020-11-22	2022-10-22	100	tablets	15	50000
33	penicillin	INJECT CARE PARENTERALS PVT LTD	2020-07-04	2022-06-04	100	antibiotics	800	6000
45	benedril	mankind	2020-06-11	2021-06-12	100	tablet	15	1500

4 rows in set (0.01 sec)

b. QUANTITY:

```
>>>
= RESTART: C:\Users\Student 06\Desktop\pharmacy management system\1672587589336_main_PMS.py
==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========
1.Add new drug details
2.delete drug details
3.update details
4.show expired drugs
5.show inventory
Enter the choice(1-5): 3
Select the name of the column to be updated
1.Drug name
2.Quantity
3.total price
Enter the choice number(1-3):2
Enter the new quantity to be updated: 75
Enter the reference number of the drug: 1
DETAILS UPDATED SUCCESFULLY
PRESS ENTER TO ENTER THE MAIN MENU
```

mysql> select * from inventory;

ref_num	DrugName	Manufturer	Manuftring_date	Expiry_date	Quantity	Category	Unit_pr	Total_pr
1	Abilify(aripiprazole)	mankind	2020-12-16	2023-04-15	75	tablet	120	20000
2	dolo	mankind	2020-11-22	2022-10-22	100	tablets	15	50000
33	penicillin	INJECT CARE PARENTERALS PVT LTD	2020-07-04	2022-06-04	100	antibiotics	800	6000
45	benedril	mankind	2020-06-11	2021-06-12	100	tablet	15	1500

4 rows in set (0.01 sec)

c. TOTAL PRICE:

```
>>> = RESTART: C:\Users\Student 06\Desktop\pharmacy management system\1672587589336_main_PMS.py
==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========
1.Add new drug details
2.delete drug details
3.update details
4.show expired drugs
5.show inventory
Enter the choice(1-5): 3
Select the name of the column to be updated
1.Drug name
2.Quantity
3.total price
Enter the choice number(1-3):3
Enter the new total price to be updated: 6000
Enter the reference number of the drug33
DETAILS UPDATED SUCCESFULLY
PRESS ENTER TO ENTER THE MAIN MENU
```

```
mysql> select * from inventory;
+---------+---------------------+--------------------------------+----------------+-------------+----------+-------------+---------+----------+
| ref_num | DrugName            | Manufturer                     | Manuftring_date | Expiry_date | Quantity | Category    | Unit_pr | Total_pr |
+---------+---------------------+--------------------------------+----------------+-------------+----------+-------------+---------+----------+
|       1 | Abilify(aripiprazole) | mankind                      | 2020-12-16     | 2023-04-15  |       64 | tablet      |     120 |    20000 |
|       2 | dolo                | mankind                        | 2020-11-22     | 2022-10-22  |      100 | tablets     |      15 |    50000 |
|      33 | penicillin          | INJECT CARE PARENTERALS PVT LTD | 2020-07-04    | 2022-06-04  |      100 | antibiotics |     800 |     6000 |
|      45 | saridon             | mankind                        | 2020-06-11     | 2021-06-12  |      100 | tablet      |      15 |     1500 |
+---------+---------------------+--------------------------------+----------------+-------------+----------+-------------+---------+----------+
4 rows in set (0.00 sec)
```

- **Show all function**

```
>>>
= RESTART: C:\Users\Student 06\Desktop\pharmacy management system\1672587589336_main_PMS.py
==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========
1.Add new drug details
2.delete drug details
3.update details
4.show expired drugs
5.show inventory
Enter the choice(1-5): 5
Ref_num:  1
Drug_name:  Abilify(aripiprazole)
Manuftr_name:  mankind
Manuftring_date:  2020-12-16
Expiry_date:  2023-04-15
Quantity:  64
Category:  tablet
Unit_pr:  120
Total_pr:  20000


PRESS ENTER TO ENTER THE MAIN MENU
```

- **Expiry function**

```
>>>
= RESTART: C:\Users\Student 06\Desktop\pharmacy management system\1672587589336_main_PMS.py
==========WELCOME TO PHARMACY MANAGEMENT SYSTEM===========
1.Add new drug details
2.delete drug details
3.update details
4.show expired drugs
5.show inventory
Enter the choice(1-5): 4
Enter todays date(YYYY-MM-DD): 2023-01-03


Ref_num:  2
Drug_name:  dolo
Manuftr_name:  mankind
Manuftring_date:  2020-11-22
Expiry_date:  2022-10-22
Quantity:  100
Category:  tablets
Unit_pr:  15
Total_pr:  50000


PRESS ENTER TO ENTER THE MAIN MENU
```

CONCLUSION

Pharmacy management system is actually a software which handle the essential data and save the data and actually about the database of a pharmacy and its management. This software helps in effectively management of the pharmaceutical store or shop. It provides the statistics about medicine or drugs which are in stocks which data can also be updated and edited. It works as per the requirement of the user and have options accordingly. It allow user to enter manufacturing as well as the expiry date of medicine placing in stock

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MADE BY:

- VAIBHAV AGARWAL (XII AT)
- RITIK GAJBHIYE (XII AP)